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PHP + MySQL

CIS 2102

DCISM

DEPARTMENT OF COMPUTER, INFORMATION SCIENCES AND MATHEMATICS

Topics

1. Introduction
2. Basics
3. Cookies and Sessions
4. Connecting MySQL to PHP

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Introduction

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PHP

- **P**ersonal **H**ome **P**age or as a recursive backronym **P**HP: **H**ypertext **P**reprocessor
- May be embedded into HTML code, or it can be used in combination with various web template systems, web content management system and web frameworks.
- A server-side scripting language that can be used to develop static websites or dynamic websites or web applications.
- Client computers accessing the PHP scripts requires only a browser.
- File extension is “.php”.

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Scripting Language

- Scripts are programming instructions that is interpreted at runtime while scripting language is a language that interprets scripts at runtime
- These are usually embedded into other software environments.
- Server-side scripts are interpreted on the server while client-side scripts are interpreted by the client application.
- PHP – server-side scripts while JavaScript – client-side script
- Both can be embedded into an HTML page.

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Programming language	Scripting language
Has all the features needed to develop complete applications.	Mostly used for routine tasks
The code has to be compiled before it can be executed	The code is usually executed without compiling
Does not need to be embedded into other languages	Is usually embedded into other software environments

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Syntax

```
<?php

        ...

?>
```

A PHP file can also contain tags such as HTML and client-side scripts such as JavaScript.

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Why PHP

- PHP is open source and free.
- Short learning curve compared to other languages such as JSP, ASP etc.
- Large community document
- Most web hosting servers support PHP by default unlike other. This makes PHP a cost-effective choice.
- PHP is regular updated to keep abreast with the latest technology trends.

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- Other benefit that you get with PHP is that it’s a server-side scripting language; this means you only need to install it on the server and client computers requesting for resources from the server do not need to have PHP installed; only a web browser would be enough.
- PHP has in built support for working hand in hand with MySQL; this doesn’t mean you can’t use PHP with other database management systems. You can still use PHP with Postgres,Oracle, MS SQL Server, ODBC etc.
- PHP is cross platform; this means you can deploy your application on a number of different operating systems such as Windows, Linux, Mac OS etc.

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Basic Output in PHP

- echo “string” or $variable or other statements;
- print “string” or $variable or other statements;
- The main difference between the print and echo statement is that print statement can have only one argument at a time and thus can print a single string. Also, print statement always returns a value 1.

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Get Started with PHP

1. Open XAMPP Control Panel and Start Apache Module.
2. Go to the install location of XAMPP and locate the htdocs folder.
3. Inside the said folder create a folder for your project.
4. Put your project files in your created project folder.
5. If you want to embed a php code into an HTML code, use the ".php" file extension instead of ".html".
6. To run your project with php code, open a browser and type in the address bar:

localhost/PROJECT_FOLDER/LOCATION_OF_YOUR_FILE

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Basics

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Datatypes

PHP is a loosely typed/weak typed language; it does not have explicit defined data types. PHP determines the data types by analyzing the attributes of data supplied.

- Alphanumeric characters are classified as **strings**
- Whole numbers are classified **integers**
  - Constant PHP_INT_MAX is used to determine the maximum value since 32-bit and 64-bit machines have different maximum values.
- Numbers with decimal points are classified as **floating points**.
- True or false values are classified as **Boolean**.

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Variable naming

- Variable names must start with the dollar sign.
- Variable names are case sensitive.
- Variable names must start with a letter follow other characters. It is illegal to start a variable with a number.
- Variable names must not contain any spaces.

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Variable type casting

- Type casting is converting a variable or value into a desired data type.
  - $b = “5”;
  - $c = $a + (int) $b;
- var_dump() function determines the data type.
  - $d = true;
  - var_dump($d);

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Constants

- A constant is a variable whose value cannot be changed at runtime.
- Defining a constant:
  - define(‘CONSTANT_NAME’, VALUE);
  - define('PI',3.14);

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Operators - Arithmetic

Operator	Name	Description	Example	Output
+	Addition	Summation of x and y	1 + 1;	2
-	Subtraction	Difference between x and y	1 – 1;	0
*	Multiplication	Multiplies x and y	3 * 7;	21
/	Division	Quotient of x and y	45 / 5;	9
%	Php Modulus	Gives reminder of diving x and y	10 % 3;	1
-n	Negation	Turns n into a negative number	-(-5);	5
x . y	Concatenation	Puts together x and y	"PHP" . " ROCKS";10 . 3;	PHP ROCKS103

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Operators - Assignment

Operator	Name	Description	Example	Output
x = ?	assignment	Assigns the value of x to ?	$x = 5;	5
x += ?	addition	Increments the value of x by ?	$x = 2;$x += 1;	3
X -= ?	subtraction	Subtracts ? from the value of x	$x = 3;$x -= 2;	1
X *=?	multiplication	Multiplies the value of x ? times	$x = 0;$x *=9;	0
X /=?	division	Quotient of x and ?	$x = 6;$x /=3;	2
X %=?	modulus	The reminder of dividing x by?	$x = 3;$x %= 2;	1
X .=?	concatenate	Puts together items	" $x = 'Pretty';$x .= ' Cool!';"	Pretty Cool!

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Operators - Comparison

Operator	Name	Example	Result
==	Equal	$x == $y	Returns true if $x is equal to $y
===	Identical	$x === $y	Returns true if $x is equal to $y, and they are of the same type
!=	Not equal	$x != $y	Returns true if $x is not equal to $y
<>	Not equal	$x <> $y	Returns true if $x is not equal to $y
!==	Not identical	$x !== $y	Returns true if $x is not equal to $y, or they are not of the same type
>	Greater than	$x > $y	Returns true if $x is greater than $y
<	Less than	$x < $y	Returns true if $x is less than $y
>=	Greater than or equal to	$x >= $y	Returns true if $x is greater than or equal to $y
<=	Less than or equal to	$x <= $y	Returns true if $x is less than or equal to $y
<=>	Spaceship	$x <=> $y	Returns an integer less than, equal to, or greater than zero, depending on if $x is less than, equal to, or greater than $y. Introduced in PHP 7.

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Operators - Logical

Operator	Name	Description	Example	Output
X and y, x && y	And	Returns true if both x and y are equal	1 and 4;True&& False;	True or 1False or 0
X or y, x || y	Or	Returns true if either x or y is true	6 or 9;0 || 0;	True or 1False or 0
X xor y	Exclusive or, xor	Returns true if only x is true or only y is true	1 xor 1;1 xor 0;	False or 0True or 1
!x	Not	Returns true if x is false and false if x is true	!0;	True or 1

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String creation

- Single quotes

If the single quote is part of the string value, it can be escaped using the backslash.

```
<?php
    echo 'I \'ll be back after 20 minutes';
?>
```

I'll be back after 20 minutes

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- Double quotes

Variable names can be used inside double quotes.

```
<?php
    $name='John';
    echo “$name is friends with Klent.”;
?>
```

John is friends with Klent.

If the $ sign is part of the string value, it can be escaped using the backslash.

```
<?php
    $pwd = “pas\$word”;
    echo $pwd;
?>
```

pas$word

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- Heredoc
  - Heredoc methodology is used to create fairly complex strings as compared to double quotes.
  - Supports all the features of double quotes and allows creating string values with more than one line without php string concatenation and using variables inside the string.
  - You can also use double quotes inside without escaping them.
  - Use <<<EOT at the beginning line of the string and just EOT at the end.
  - EOT means “end of text”

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```
<?php
    $name = “Chris”;
    echo <<<EOT
        I am $name.
        You can call me “Chiz”.
    EOT;
?>
```

I am Chris. You can call me “Chiz”.

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- Nowdoc
  - Similar to heredoc but works like a single quote.
  - Use <<<'EOT' at the beginning line of the string and just EOT at the end.

```
<?php
    $name = "Chris";
    $message = <<<'EOT'
        I am $name.
        You can call me "Chiz".
    EOT;
    echo $message;
?>
```

I am $name. You can call me "Chiz".

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Common string functions

Function	Description	Example	Output
strtolower	Used to convert all string characters to lower case letters	echo strtolower('Benjamin');	outputs benjamin
strtoupper	Used to convert all string characters to upper case letters	echo strtoupper('george w bush');	outputs GEORGE W BUSH
strlen	The string length function is used to count the number of character in a string. Spaces in between characters are also counted	echo strlen('united states of america');	24
explode	Used to convert strings into an array variable	$settings = explode(';', "host=localhost; db=sales; uid=root; pwd=demo"); print_r($settings);	Array ([0] => host=localhost [1] => db=sales [2] => uid=root [3] => pwd=demo)
substr	Used to return part of the string. It accepts three (3) basic parameters. The first one is the string to be shortened, the second parameter is the position of the starting point, and the third parameter is the number of characters to be returned.	$my_var = 'This is a really long sentence that I wish to cut short';echo substr($my_var,0, 12).'...';	This is a re...

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str_replace	Used to locate and replace specified string values in a given string. The function accepts three arguments. The first argument is the text to be replaced, the second argument is the replacement text and the third argument is the text that is analyzed.	echo str_replace ('the', 'that', 'the laptop is very expensive');	that laptop is very expensive
strpos	Used to locate the and return the position of a character(s) within a string. This function accepts two arguments	echo strpos('PHP Programing','Pro');	4
sha1	Used to calculate the SHA-1 hash of a string value	echo sha1('password');	5baa61e4c 9b93f3f0 682250b6cf8331b 7ee68fd8
md5	Used to calculate the md5 hash of a string value	echo md5('password');	9f961034ee 4de758 baf4de09ceeb1a75
str_word_count	Used to count the number of words in a string.	echo str_word_count ('This is a really long sentence that I wish to cut short');	12
ucfirst	Make the first character of a string value upper case	echo ucfirst('respect');	Outputs Respect
lcfirst	Make the first character of a string value lower case	echo lcfirst('RESPECT');	Outputs rESPECT

Explore and try each string function. Go to: https://www.php.net/manual/en/ref.strings.php for more string functions.

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Control Structures

- Categorization of code execution
  - Sequential – execution of code in order
  - Decision – making a choice
  - Repetition – loops

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Decision

- if

```
if (condition) {
    …
}
```

- if-else

```
if (condition) {
    …
} else {
    …
}
```

- if-elseif-else

```
if (condition1) {
    …
} elseif (condition2) {
    …
} else {
    …
}
```

- Nested if-else

```
if (condition1) {
    if (condition1.1) {
        …
    } else {
        …
    }
} else {
    if (condition2.1) {
        …
    } else {
        …
    }
}
```

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- switch-case

```
switch (condition) {
    case 1:
            …
            break;
    case 2:
            …
            break;
    case 3:
            …
            break;
    default:
            …
            break;
}
```

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Repetition

- For-loop

```
for (initialization; condition; increment/decrement) {
        …
}
```

- While-loop

```
while(condition) {
    …
}
```

- For-each-loop

```
foreach ($array_variable as $array_values) {
    …
}
```

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Arrays

is a variable that stores more than one piece of homogenous data in a single variable.

Initialization:

- $emptyArray = [];
- $emptyArray = array();
- $emptyArray = (array) null;
- $array = array("Gundam","Naruto","Dragon Ball Z","Bleach","Voltes V");
- $array = array(0 => "Gundam",
  1 => "Naruto",
  2 => "Dragon Ball Z",
  3 => "Bleach",
  4 => "Voltes V");

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Numeric array

- use number as access keys.
- $variable_name[n] = value;

Associative array

- use descriptive names for id keys.
- $variable_name[“key_name”] = value;
- $variable_name = array(“keyname” => value);

```
$name = array("first_name" => "John",
        "last_name" => "Doe",
        "middle_name" => "Smith");
echo “Name: $name[“last_name”], $name[“first_name”]“;
```

Name: Doe, John

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Multi-dimensional array

- Contain other nested arrays.
- $array = array(“key1” => array(“value1”,”value2”,”value3”),
  “key2” => array(“value1”,”value2”,”value3”),
  “key3” => array(“value1”,”value2”,”value3”));

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- $array = array(“key1” => array(
  0 => “value1”,
  1 => ”value2”,
  2 => ”value3”),
  “key2” => array(
  0 => “value1”,
  1 => ”value2”,
  2 => ”value3”),
  “key3” => array(
  0 => “value1”,
  1 => ”value2”,
  2 => ”value3”));
- echo $array[“key1”][0]

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Common array function

- count($array) -> Counts the number of elements in the array. Returns an integer.
- is_array($array) -> Used to determine if variable is an array or not. Returns a bool.
- sort($array) -> Sorts array by values. Returns a numeric array.
- ksort($array) -> Used for associative array. Sorts it by key.
- asort($array) -> Used for associative array. Sorts it by value.
- array_push($array, value1, value2, ...) -> Used to insert values in an array.
- array_splice($array, start, length) -> Removes selected elements from an array and replaces it with new elements.
    - start – Required numeric value that indicates the position where the removal starts.
    - length – Optional numeric value that indicates how many items to be removed in an array.

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Functions

- Functions are simple code blocks we can call from anywhere in our code.
- This can either be built-in functions or user-defined functions.

Syntax:

- User-defined function:

```
function functionName(parameters){
        …
        return $return_value; //optional
}
```

- Function call:
  - functionName(arguments);
  - $catcherVariable = functionName(arguments);

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Cookies and Sessions



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Cookies

- It is a small file with the maximum size of 4KB that the web server stores on the client computer.
- A cookie can only be read from the domain that it has been issued from.
- A cookie created by a user/client can only be visible to them. It can also be disabled.

How it works:

- A user requests for a page that stores cookies
- The server sets the cookie on the user's computer
- Other page requests from the user will return the cookie name and value

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Syntax:

setcookie(cookie_name, cookie_value, [expiry_time], [cookie_path], [domain], [secure], [httponly]);

PHP set cookie function must be executed before the HTML opening tag.

- cookie_name is the name of the cookie that the server will use when retrieving its value from the $_COOKIE array variable.
- cookie_value is the value of the cookie.
- [expiry_time] can be used to set the expiry time for the cookie such as 1 hour. The time is set using the PHP time() functions plus or minus a number of seconds greater than 0 i.e. time() + 3600 for 1 hour. Default expiration is 30 days.
- [cookie_path] can be used to set the cookie path on the server. The forward slash “/” means that the cookie will be made available on the entire domain. Sub directories limit the cookie access to the subdomain.

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- [domain] can be used to define the cookie access hierarchy i.e. www.cookiedomain.com means entire domain while www.sub.cookiedomain.com limits the cookie access to www.sub.cookiedomain.com and its sub domains.
- [secure] is used to determine whether the cookie is sent via https if it is set to true or http if it is set to false. Default value is false.
- [Httponly] if set to true, then only client-side scripting languages i.e. JavaScript cannot access them.

Sample:

```
<?php
    setcookie(“user_name”, “cbelarmino”, time()+ 60,”/”);
    // expires after 60 seconds
?>
```

Accessing:

```
echo $_COOKIE[“user_name”];
```

To delete a cookie, set the expiry time to a time that has already passed or time() -360.

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Sessions

- A session is a global variable stored on the server.
- Each session is assigned a unique id which is used to retrieve stored values.
- Whenever a session is created, a cookie containing the unique session id is stored on the user's computer and returned with every request to the server. If the client browser does not support cookies, the unique php session id is displayed in the URL
- Sessions have the capacity to store relatively large data compared to cookies.

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- The session values are automatically deleted when the browser is closed. If you want to store the values permanently, then you should store them in the database.
- Just like the $_COOKIE array variable, session variables are stored in the $_SESSION array variable. Just like cookies, the session must be started before any HTML tags.
- Session can only store values with primary data types.

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Setting sessions

- session_start(); - starts the PHP session
- $_SESSION[“key”] = value; - directly assign a value like an array.
- Check if session exists:

```
if(isset($_SESSION[“key”]))
{
        …
}
```

- session_destroy(); - destroy the entire session.
- unset($_SESSION[“key”]); - destroys a specific session item.

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Connecting MySQL to PHP



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$con = mysqli_connect($host, $user_name, $password, $db_name);

- Used to connect to a MySQL database server. Returns a database object connection resource stored in a variable.
- $host is the name or IP address of the server hosting MySQL server. In our case, localhost.
- $user_name is a valid user name in MySQL server. Sample is root.
- $password is a valid password associated with a user name in MySQL server. By default, if you are using root, then the password is just an empty string.
- $db_name is the name of your MySQL database.

$sqlCreateDBQuery = “CREATE DATABASE database_name”;

- Creates a MySQL database given a database_name.

$sqlCreateTableQuery = “CREATE TABLE table_name(table_fields)”;

- Creates a MySQL table given a table _name and table_fields.

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```
$con->query($sqlCreateDBQuery);

$con->query($sqlCreateTableQuery);
```

- Executes the query.
- Returns True if successful, otherwise false.

```
$stmt = $con->prepare($query);

$stmt->bind_param("s", $param);

$stmt->execute();
```

- $query – db query
- prepare() – prepares your db query
- bind_param(“s”,$param) – binds the paramaters included in your db query
- execute() – executes your db query

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$result = $stmt->get_result();

- get_result() – gets and return the result depending on your query.

$stmt->close();

- closes connection to the MySQL db server.

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